

Nishtha Gupta

Backend & Cloud Engineer | AWS Certified Solutions Architect

nishtha.gupta.446@gmail.com | College Park, MD | LinkedIn | GitHub | Portfolio

EDUCATION

- University of Maryland, College Park** College Park, MD
• *Master of Engineering in Cloud Engineering* Jan 2026 – Dec 2027 (Expected)
 - **Coursework:** Cloud Computing, Advanced Networking, Databases, Virtualization and Container Technology
- University of Mumbai** Mumbai, India
• *Bachelor of Engineering in Computer Science and Engineering* Aug 2019 – May 2023

WORK EXPERIENCE

- TIAA** Mumbai, India
• *Software Analyst – Best Fresher of the Year (2024)* Jul 2023 – Nov 2025
 - **Production Debugging:** Diagnosed customer-facing failures across MuleSoft APIs, Master Data Management (MDM), Snowflake, and Salesforce from logs, API traces, and backend workflows, giving developers root causes.
 - **API & Data Integrity:** Validated MuleSoft API data loads and reconciled Salesforce against Snowflake with SQL and Java scripts, catching 5–7 critical issues per release; investigated MDM duplicate-profile defects, cutting escalations 60%.
 - **Legacy Platform Migration:** Audited the Siebel-to-Salesforce migration across millions of records and 75+ legacy entities, surfacing 150+ field-level discrepancies before production cutover.
 - **CI/CD Automation:** Automated 100+ integration and regression tests in Java (TestNG) within Copado CI/CD, cutting execution time from 2 hours to under 15 minutes and recovering 100+ engineering hours per sprint.
- SHRAM (B2B Productivity Platform)** Mumbai, India
• *Backend Team Lead (Part-time; promoted from Backend Developer)* Nov 2021 – May 2022; Aug 2022 – Jan 2023
 - **Backend Migration:** Migrated the backend module by module from Java/Spring Boot to Node.js while the legacy system stayed live, preserving REST API contracts and executing a controlled ~2-hour production cutover.
 - **Query Optimization:** Built task-assignment and delegation APIs in Node.js, Express.js, and MariaDB/MySQL; indexed lookup paths and consolidated repeated queries, cutting database round trips from up to 3 per request to 1.
 - **Analytics Service:** Built an analytics service and REST API layer for web dashboards, improving load times by 30%.
 - **Team Leadership:** Led a 3-engineer backend team serving 15–20 customer teams and owned deployments and production fixes; set API design and code-review standards that reduced frontend/backend integration rework.
 - **Notifications & Security:** Integrated Firebase notifications for web and mobile clients (~70 daily, ~250 peak); implemented authentication, authorization, input validation, and data encryption.
- TIAA** Pune, India
• *Student Intern (Software Engineering)* Jun 2022 – Jul 2022
 - **Backend Development:** Built Python REST APIs with Django REST Framework for an AI/NLP application.

PROJECTS

- Lottery Verification Platform** – Infrastructure & Networking Lead (AWS, Terraform, ECS, SQS, Redis) (2026)
 - Designed a multi-AZ VPC with an ALB, ECS Fargate services, and port-scoped security groups, provisioned and torn down end to end via 10 modular Terraform components.
 - Decoupled claim processing from the verification path with SQS and a dead-letter queue (maxReceiveCount=3, 14-day retention), keeping verification available during claims-service failures.
 - Cut p99 verification latency from 80–150 ms to 2–10 ms with an ElastiCache Redis cache-aside layer, benchmarking cold and warm paths via custom CloudWatch metrics (CacheHit, CacheMiss, VerificationLatency).
 - Excluded WINNER outcomes from caching to prevent stale claim state; kept a shared RDS instance over database-per-service, with SQS payloads carrying enough context for the claims service to run independently.
- AWS 3-Tier Infrastructure Platform (Academic)** (Terraform, EC2, ALB, Auto Scaling, RDS) (2026)
 - Codified an ALB → Auto Scaling web tier → RDS stack in 5 Terraform modules, isolating databases in private subnets behind a NAT Gateway, with environment-aware resource counts serving dev, staging, and prod from one codebase.
- Surge** – Message-Driven Kubernetes Autoscaling (Academic) (AWS EKS, KEDA, SNS, SQS, Helm) (2026)
 - Built an SNS → SQS → EKS worker pipeline with a KEDA ScaledObject scaling pods from 1 to 10 on queue depth; provisioned clusters via eksctl and Helm with least-privilege IAM policies.
 - Validated scale-out and scale-in under a 100-message synthetic load test using CloudWatch Container Insights.
- Polyglot Healthcare Management System** (Python, PostgreSQL, MongoDB, Neo4j, Docker) (2026)
 - Built the Python repository layer over PostgreSQL and a Neo4j graph of diseases, medications, symptoms, and lab tests, running traversal-based drug-interaction checks that gate prescriptions before they are recorded.

TECHNICAL SKILLS

- **Languages:** Python, Java, JavaScript, SQL, Bash
- **Backend:** REST APIs, Node.js, Express.js, Django REST Framework, Spring Boot, event-driven architecture, caching
- **Cloud & Infra:** AWS (ECS Fargate, EKS, EC2, VPC, ALB, RDS, SQS, ElastiCache, IAM), Terraform, Docker, Kubernetes
- **Databases:** PostgreSQL, MySQL/MariaDB, Redis, MongoDB, Neo4j, Snowflake
- **DevOps & Observability:** Git, CI/CD, Copado, Helm, KEDA, CloudWatch, Splunk, Linux

CERTIFICATIONS

- AWS Certified Solutions Architect – Associate | HashiCorp Terraform Associate | AWS Certified AI Practitioner